

Kanav Bengani

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WORK EXPERIENCE

- HubSpot | *Software Engineer* December 2025 – Present
- Built **agentic search** to enhance discovery of **6,000+** published Marketplace integrations using MCP tools and skills
 - Implemented a self-serve platform serving **80K+** users/week allowing marketing teams to update site content with no engineering support; designed SQL-backed database models, low-latency RPCs, and React UI components
 - Resolved customer queries, high-severity incidents, and technical alerts (up to **40/week**) on a 24/7 on-call rotation
- Capital One | *Software Engineering Intern* June 2025 – August 2025
- Assessed image quality for over **100K** documents/day (i.e. government-issued IDs, W-2s, and bank statements)
 - Extracted image data with **85%** accuracy using **OCR** and surfaced it on a web app created using Flask and React
- Klaviyo | *Software Engineering Co-op* January 2025 – May 2025
- Optimized nightly segmentation jobs to be **7x** faster by ingesting **91%** less duplicate records per second
 - Utilized **Django** (backend), **ClickHouse** (sharded database), and **Kafka** (event streams) to process customer queries
 - Distributed deployment across **192 Kubernetes** pods and managed cluster configurations using **ZooKeeper**
- Fidelity Investments | *Software Engineering Intern* June 2024 – August 2024
- Automated the archival of approximately **500** internal associate compliance alerts per day for auditing purposes
 - Used **Azure Functions** to create APIs that retrieve alert data, package into a pdf, and send to document storage
- Johnson & Johnson | *Machine Learning Co-op* January 2024 – May 2024
- Developed **71%** accurate **scikit-learn** models in **Python** to help reps identify patients unlikely to receive therapy
 - Compared clustering (k-means/hierarchical), random forest, and decision tree as possible solutions

PROJECTS

- Ballboy | Fantasy Tennis Mobile App
- Founded a platform launched on **iOS** with **1.5K+** impressions where users join fantasy leagues following live tennis
 - Led engineering using tools like Django, Redis, React Native, and Cloudflare to ensure a **low-latency, real-time** app
 - Achieved **100K+** interactions while building a social media following across platforms (i.e. Instagram, TikTok)
- Invariance | AI-governed Dynamic Site Customization
- Architected an **SDK** that allows live UI restyling and restricted backend updates with a plain-English request per user
 - Looped LLM generation against an automated verifier and **LLM-as-a-judge** until they clear developer-set invariants
 - Devised a developer console to track website modifications analytics, conflict resolutions, and end-user **versioning**
- Reddit Sentiment Analysis | Agentic Chatbot
- **Fine-tuned** an LLM that constructs sentiment from the respective subreddit, given a natural language query
 - Parsed and filtered context using **RAG** and a BERT classifier; derived sentiment using the Meta Llama 3.1 8B model

RESEARCH

- FactEvo Optimization
- Utilized **evolutionary computing** for a course scheduling optimization problem resulting in pareto-optimal solutions
- Shriners Children's Hospital Biomed AI
- Co-authored a **publication** improving surgical outcomes for cerebral palsy patients with a predictive **autoencoder**

EDUCATION

Northeastern University, Boston, Massachusetts **3.95/4.0**
B.S. Major in **Data Science**, Minor in **Mathematics** May 2025
Coursework: Algorithms, Object-Oriented Design, LLM-Engineered Systems, Database Design, Artificial Intelligence, NLP, ML 1 + 2

SKILLS

Programming Languages: Python, Java, SQL, C/C++, JavaScript, HTML, CSS
Frameworks: React, Flutter, Django, Flask, Tailwind, NumPy, Pandas, Scikit-learn, Matplotlib, PyTorch
Tools: AWS, Kubernetes, Docker, Jenkins, ZooKeeper, Git, MongoDB, ClickHouse, Redis, Kafka, PostgreSQL, Grafana